THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE UNIT TEST SET 1

Chapters 1–4: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation, The Atomic Foundations of Matter, Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. In science, a model is best described as: 1

(a) a final truth that never changes
(b) an exact copy of the real object
(c) an untested guess
(d) a simplified representation of a system that keeps what matters for the question

2. A pure substance consists of: 1

(a) only metals
(b) only gases
(c) a single kind of particle with fixed composition
(d) two or more substances in any ratio

3. The law of conservation of mass was proposed by: 1

(a) J. J. Thomson **(b)** Joseph Proust
(c) John Dalton **(d)** Antoine Lavoisier

4. The sub-atomic particle discovered by J. J. Thomson was the: 1

(a) electron **(b)** proton **(c)** nucleus **(d)** neutron

5. A rough estimate is useful when: 1

(a) an approximate answer is enough for a decision
(b) an exact value is required
(c) we want to avoid reasoning
(d) no data are available at all

6. A suspension is: 1

(a) a clear solution
(b) a compound
(c) a pure substance
(d) a heterogeneous mixture whose particles settle on standing

7. The atomicity of phosphorus (P_4) is: 1

(a) 8 **(b)** 1 **(c)** 4 **(d)** 2

8. The atomic number of an element is equal to the number of: 1

(a) valence electrons only **(b)** neutrons
(c) nucleons **(d)** protons

9. A 'law' of science and a 'theory' differ because: 1

(a) a law describes what happens while a theory explains why
(b) they are the same
(c) a law is a guess
(d) a theory is untested

10. In paper chromatography the pencil line is drawn at the bottom because: 1

(a) ink is better **(b)** pencil does not dissolve in the solvent
(c) it looks neat **(d)** pencil is faster

11. An anion is a: 1

(a) negatively charged ion **(b)** neutral particle
(c) molecule **(d)** positively charged ion

12. The valency of an element with 7 valence electrons is: 1

(a) 1 **(b)** 6 **(c)** 7 **(d)** 8

13. Chemists draw atoms as spheres and bonds as sticks. This is an example of: 1

(a) a law **(b)** a model **(c)** an estimate **(d)** a unit

14. Which of the following is a pure substance? 1

(a) distilled water **(b)** air **(c)** sea water **(d)** milk

15. The law of constant proportions was proposed by: 1

(a) Joseph Proust **(b)** Lavoisier **(c)** Rutherford **(d)** Dalton

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. What is a model in science? Give one example from chemistry. 2

17. Differentiate between a homogeneous and a heterogeneous mixture with one example each. 2

18. State the law of conservation of mass. Who proposed it? 2

19. State two limitations of Thomson's model of the atom. 2

20. Why is ignoring air resistance not a mistake when studying a falling ball? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain with an example how a model is built and why assumptions are made. 3

22. Explain the method of separating a mixture of salt, sand and ammonium chloride. 3

23. A reaction between 6.0 g of carbon and 16.0 g of oxygen gives CO2. Verify the law of conservation of mass. 3

24. Describe the gold foil experiment: set-up, observations and the conclusion. 3

OR

How does the incident of the aircraft running out of fuel show the importance of units? 3

25. How much solute is needed to prepare 250 g of a 12 per cent (m/m) sugar solution? 3

OR

State three properties of ionic compounds. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain the characteristics of the scientific method with an example of an investigation, including question, hypothesis, experiment, observation and conclusion. 5

OR

Differentiate between pure substances and mixtures. Classify each as pure, homogeneous mixture or heterogeneous mixture: air, sugar solution, gold, milk, brass. 5

27. Describe an experiment to verify the law of conservation of mass using sodium sulphate and barium chloride solutions in a closed flask. State the observation and conclusion. 5

OR

Describe Thomson's, Rutherford's and Bohr's models of the atom with one drawback of each of the first two. 5

END OF QUESTION PAPER Total: 50 Marks

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